

IBM University Engagement Project Submission

*“Exploring the Power of Agentic AI with IBM watsonx and IBM watsonx
orchestrate ”*

Project Title – [Eco Lifestyle Agent: A Multi-Agent RAG Framework for Sustainable Living]

Domain of Project – [AI for Sustainability & Environmental Conservation (GreenTech)]

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Omanshu Rawat	 A passport-sized photograph of a young woman with dark hair, wearing a red sweater over a white collared shirt. Below the photo is a small white label with the text 'OMANSHU RAWAT' and '16-02-2026'.	omanshurawat27@gmail.com	7067853866

Problem Statement -

Problem Statement : Eco Lifestyle Agent

The Challenge

Adopting a sustainable lifestyle is crucial for combatting climate change, but everyday users face fragmented, confusing, and conflicting information regarding recycling policies, eco-friendly product swaps, and green habits. Local waste management protocols are highly specific, and general internet searches often fail to provide localized, actionable guidance. Without a centralized, trustworthy platform that provides clear, personalized, and real-time advice, individuals struggle to transition from eco-awareness to actual eco-action.

The Objective

Build an AI-powered Eco Lifestyle Agent that delivers personalized, practical sustainability insights to empower users to adopt greener habits. The solution should:

Localized Waste Management & RAG Retrieval – Fetch accurate, region-specific recycling rules, composting methods, and waste segregation strategies from verified environmental datasets.

Sustainable Product & Habit Alternative Guidance – Recommend eco-friendly alternatives to single-use plastics and high-carbon household products based on user needs.

Actionable Carbon Footprint Reduction – Break down complex environmental milestones into small, high-impact daily habits that individuals can easily practice.

Multi-Agent Collaborative Architecture – Leverage specialized conversational and processing blocks to separate core retrieval, validation, and interaction tasks.

Technology Used

Orchestration Engine: IBM watsonx Orchestrate

Foundation Models: GPT-OSS 120B

Data Strategy: Retrieval-Augmented Generation (RAG)

Problem Statement - Tech Stack (Mandatory)

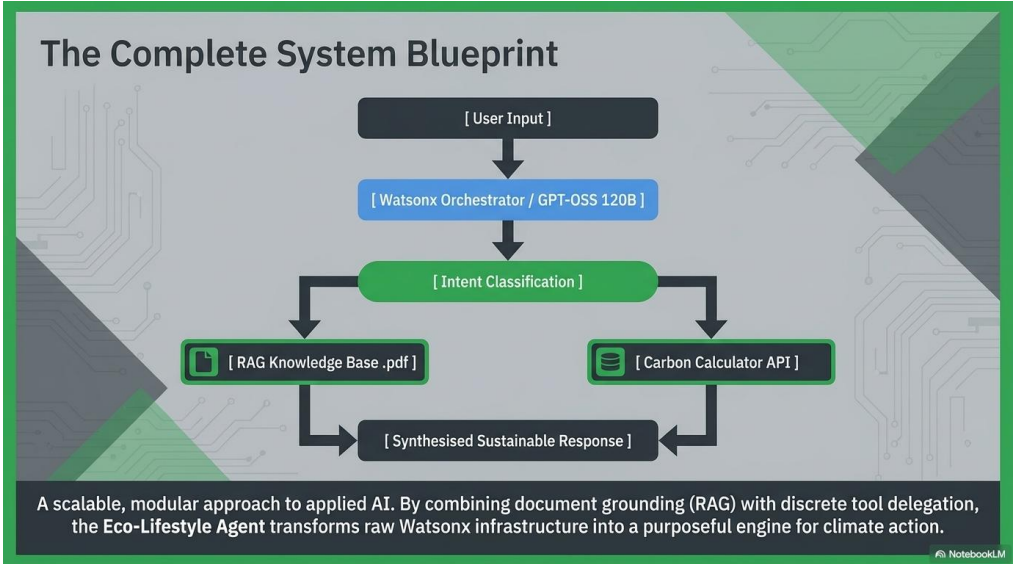
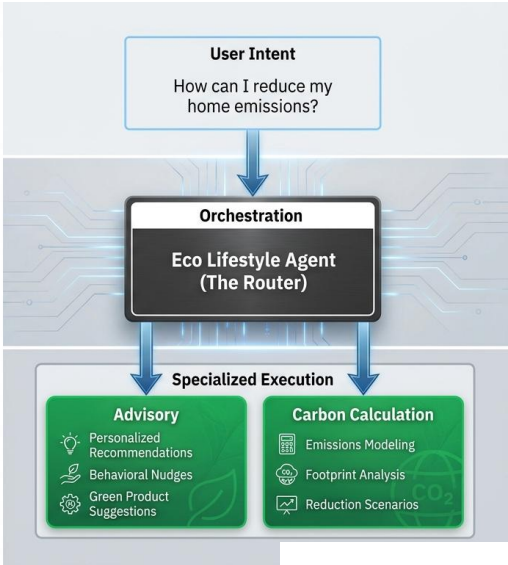
Approach Used: IBM watsonx orchestrate + IBM watsonx.ai

Proposed Solution - This project proposes an AI-driven, multi-agent assistant that empowers users to adopt sustainable habits through a personalized, hallucination-free RAG knowledge pipeline

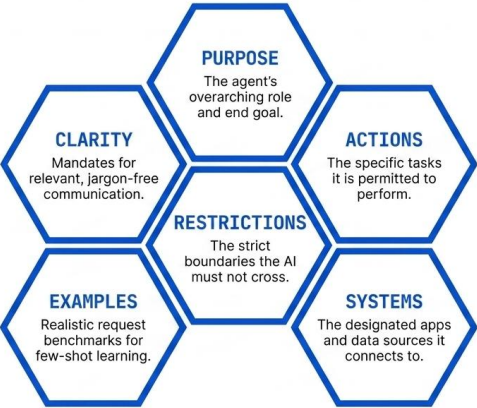
Intelligent Orchestration for Sustainable Action

Rather than relying on a single, monolithic language model, the Eco-Lifestyle Agent operates as an intelligent router. It dynamically assesses user queries regarding environmental impact and delegates tasks to specialized sub-agents, grounded in curated sustainability data.

BUILT FOR PRECISION, POWERED BY IBM WATSONX ORCHESTRATE.



The Agent Definition Framework



Create an agent

Name *

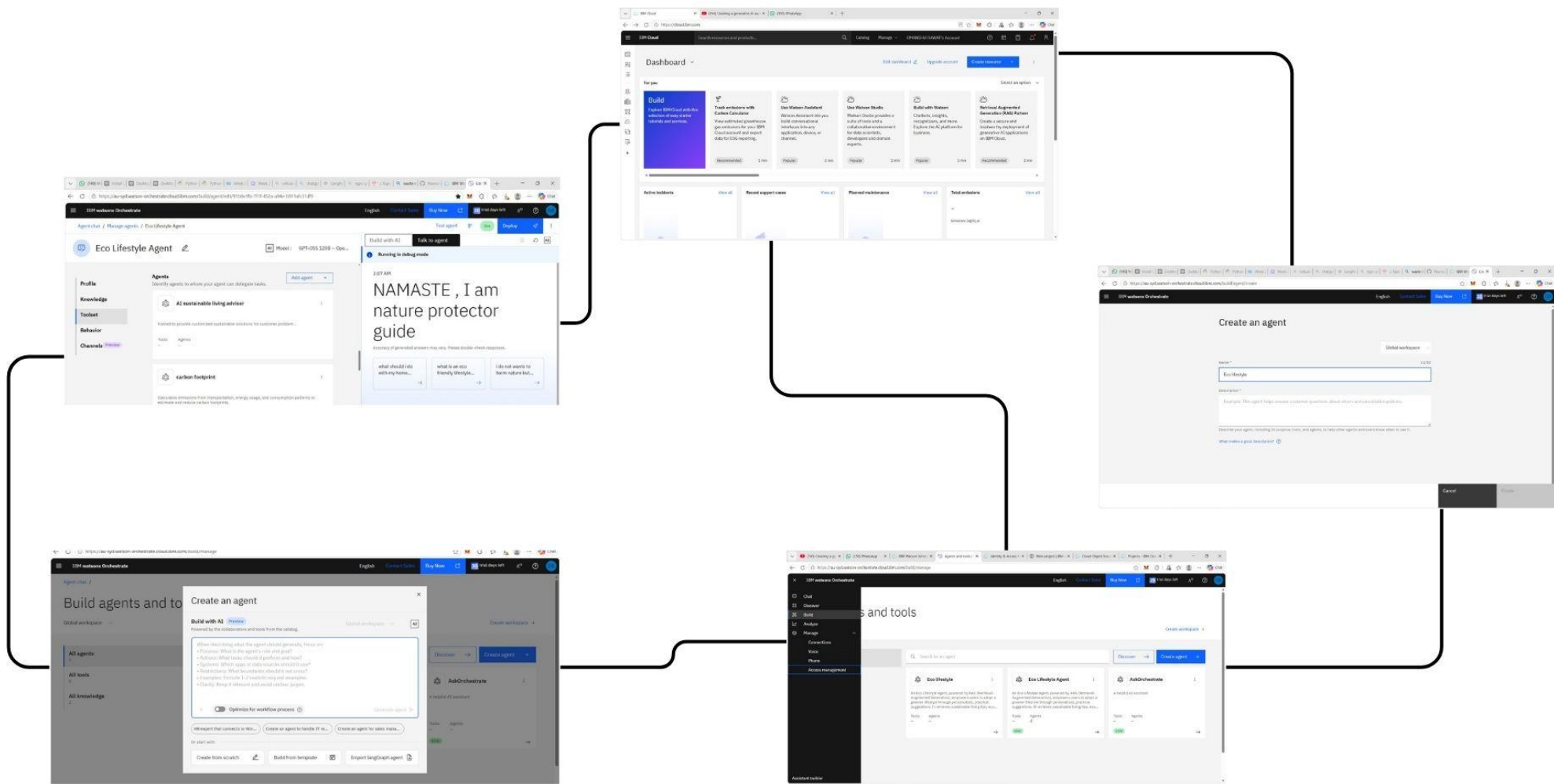
Eco lifestyle

Description *

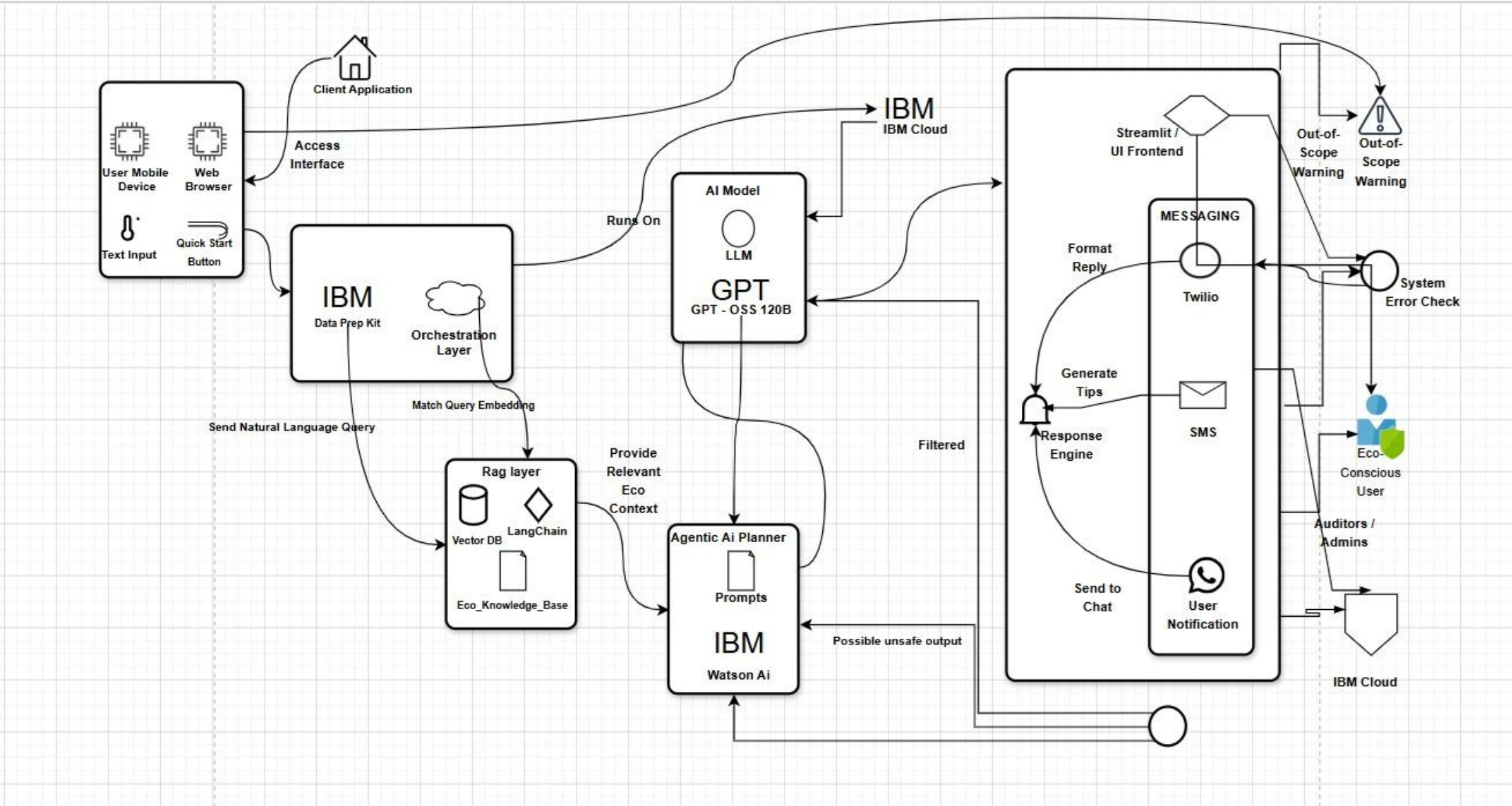
Orchestrate component Used -

- 1) **Chat Input** – The interactive messaging UI where users submit eco-queries such as *"What should I do with my home generated waste?"* or *"What is an eco friendly lifestyle looks like?"*
- 2) **IBM watsonx AI Agent** - The processing core that parses user inputs using custom prompt instructions and matches queries to appropriate tools or knowledge stores.
- 3) **Name of IBM GPT-OSS 120B model** - **GPT-OSS 120B** is a high-capacity, open-source foundation model integrated into your workspace to handle the complex reasoning, custom tool routing, and multi-turn chat interactions for your agents.
- 4) **Chat output** -The visual interface rendering clear, encouraging, structured markdown bullet points containing actionable green tips.
- 5) **File component** - Used during the ingestion phase to parse and chunk the custom text file (eco_friendly_lifestyle_knowledge_base.pdf) containing the primary sustainability corpus.

IBM watsonx Orchestrate workflow -



Architecture blueprint



Role of Agentic AI in the solution

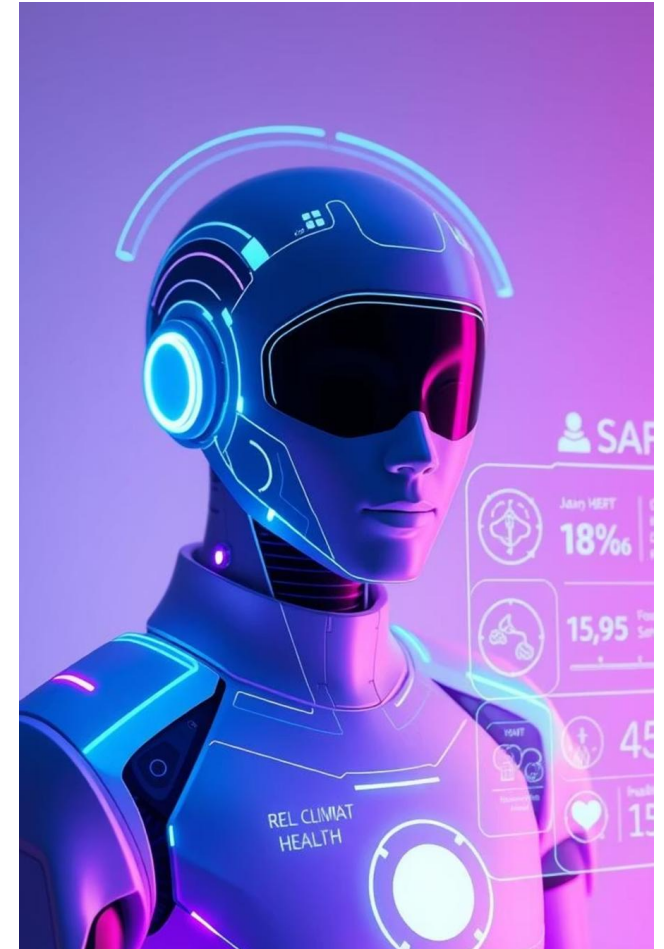
Role of Agentic AI in Eco lifestyle Agent

In **Eco Lifestyle** project, Agentic AI elevates the system from a basic keyword-matching search tool into an **autonomous, goal-oriented sustainability consultant**..

Dynamic Intent Routing: Instead of just generating text, the agent acts as an automated decision-maker. It analyzes the user's natural language input to determine whether to fetch answers from your local RAG data asset (eco friendly lifestyle module) or delegate complex tasks to specialized tools like the carbon footprint sub-agent.

Autonomous Step-by-Step Planning: When a user asks a broad question like *"How can I reduce waste at home?"*, Agentic AI uses its reasoning loop to break the goal down into sequential phases (e.g., teaching waste segregation first, suggesting upcycling alternatives second, and routing to local collection programs third) to create an actionable roadmap.

Contextual Problem Solving: It adapts to user readiness by offering practical lifestyle modifications based on real-time feedback, ensuring that the guidance delivered is hyper-personalized, realistic, and focused on driving quantifiable environmental conservation.



Technology Used

- **IBM watsonx orchestrate** -For designing and orchestrating multi-agent workflows visually
- **GPT-OSS 120B model** - For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

Novelty and Uniqueness

The proposed Nutrition Agent stands out by combining **Agentic AI, RAG, and personalization** into a single intelligent system.

Multi-Agent Intelligence – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate and context-aware nutrition guidance.

RAG-Based Accuracy – Integrates real-time data retrieval with AI generation, reducing misinformation and ensuring trustworthy recommendations.

Hyper-Personalization – Tailors diet plans based on health conditions, cultural preferences, and fitness goals.

Real-Time Feedback Loop – Continuously adapts recommendations based on daily food logs and user behavior.

Multi-Modal Input – Supports text, voice, and image-based meal tracking for better usability.

Preventive Healthcare Focus – Goes beyond diet planning to provide disease-specific nutrition advice.

Scalable & Enterprise-Ready – Built on **IBM watsonx.ai** with **GPT-OSS 120B**, ensuring reliability and governance.

This makes the solution a **smart, adaptive, and proactive digital nutritionist**, not just a static diet recommendation tool.

Git Hub Link

Working GitHub Repository URL: [https://github.com/Neerura/Eco-Lifestyle-Agent-Watsonx]

(<https://github.com/Neerura/Eco-Lifestyle-Agent-Watsonx>)

Make sure you have Uploaded following three files into your github repository

1) app.json file , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4)any other files

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Future Scope

1) Direct Scrap-Tech & Circular Startup Integrations –

The primary future milestone is establishing direct API connectivity with growing scrap collection, waste management, and upcycling startups. The platform will dynamically connect users directly to active field networks such as:

The Kabadiwala & Scrap Uncle: To allow users to instantly book door-to-door dry waste and scrap collections directly from the chat interface.

Nokasa: To facilitate structured plastic collection drives and track traceable automated trash disposal drop-offs.

Regional Clothing Recommerce Hubs: Integrating with progressive textile startups (such as circular apparel platforms in Bangalore) to trade or monetize old, worn clothing rather than letting them enter municipal landfills.

1) Gamified Green Badges and Dynamic Carbon Metrics

Introduce a micro-tracking framework where users can input completed daily tasks (e.g., *"I used a cloth bag today"*). The agent will calculate approximate plastic/carbon offset metrics and reward users with unlockable digital badges to gamify the journey toward low-impact sustainable living.

Thank You!

Thank you for your time and interest.